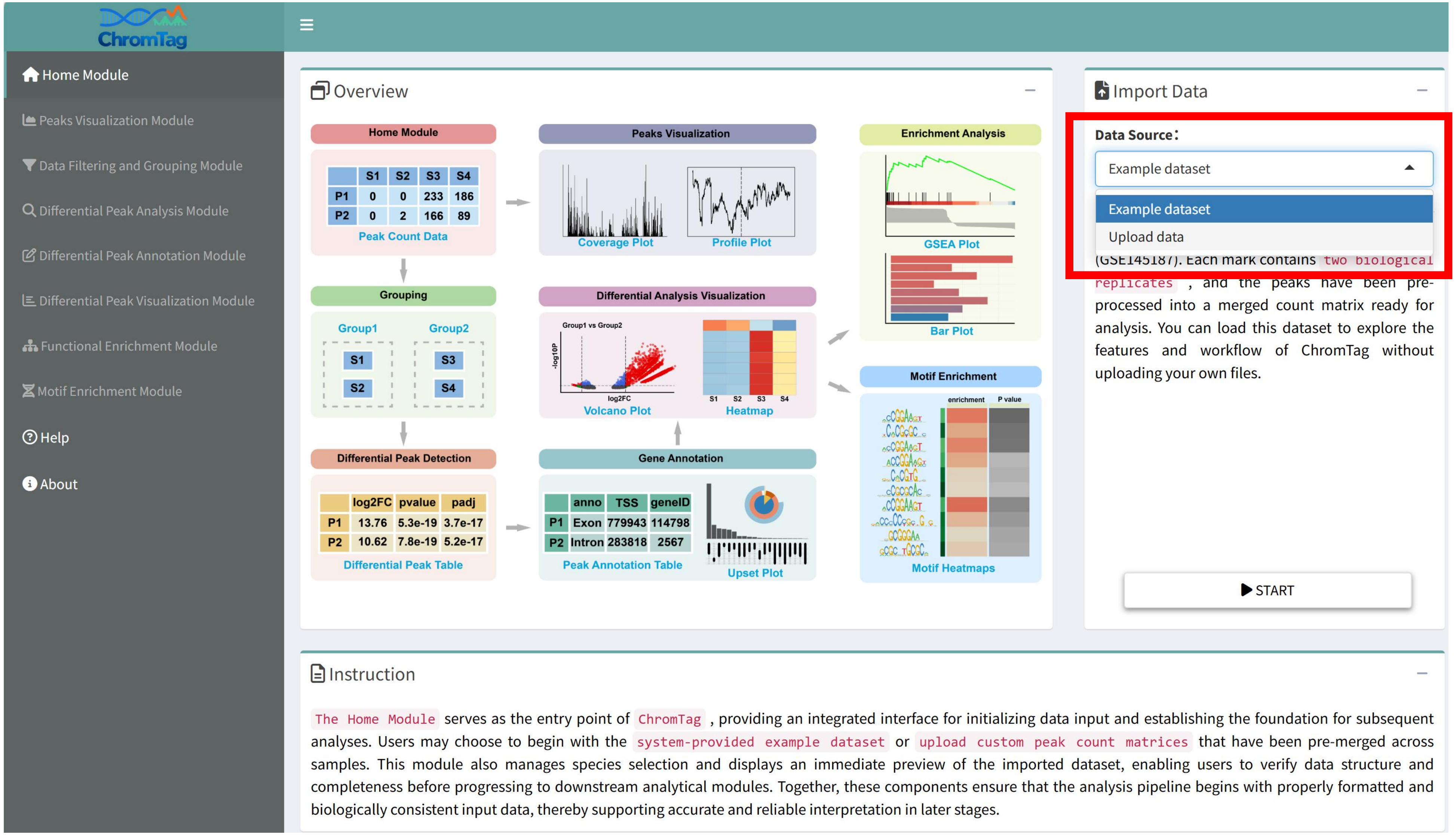
ChromTag

Basic Tutorial

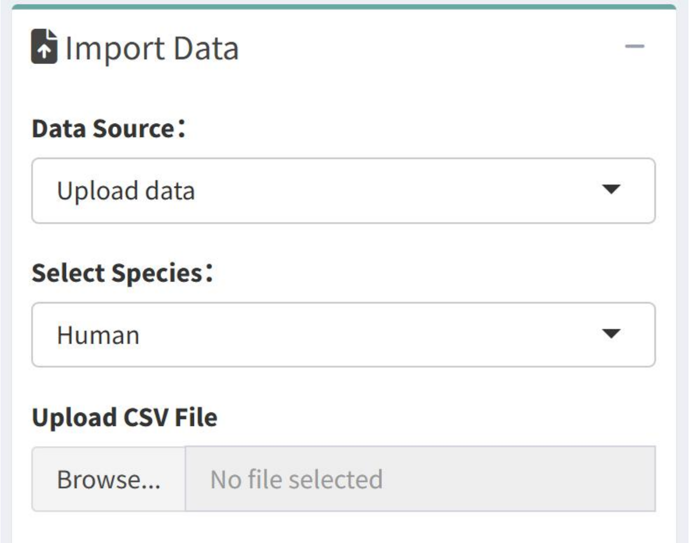
Home Module



Users can choose between **“Example Dataset”** and **“Upload Data”**.

If the **Example Dataset** is selected, the application automatically loads a **pre-merged peak count matrix** containing H3K27me3 and H3K4me3 samples, each with two biological replicates.

Home Module



If you choose to upload custom data, you must also select the appropriate species.

	Chromosome	Start	End	H3K27me3_rep1	H3K27me3_rep2	H3K4me3_rep1	H3K4me3_rep2
2	chr1	27805	30656	0	0	233	186
3	chr1	135727	140083	0	2	166	89
4	chr1	198335	200968	0	0	219	169
5	chr1	391014	392146	1	3	0	1
6	chr1	392617	393317	2	1	0	0
7	chr1	393808	394451	2	1	0	0
8	chr1	440860	441530	4	2	0	1
9	chr1	442100	442830	2	0	0	0
10	chr1	492154	494761	0	1	73	36

Uploaded data should be **pre-merged across samples** and must follow a specific format. The file should be a **CSV** with a header row.

The first column must contain the chromosome name (e.g., “chr1”), the second and third columns should specify the **start** and **end** positions, and all subsequent columns should represent the **peak count values** for each sample.

Each experimental group must include **at least two biological replicates** to ensure proper statistical analysis.

Home Module

Import Data

Data Source:

Example dataset

- Example dataset
- Upload data

(GSE145187). Each mark contains `two biological replicates` , and the peaks have been pre-processed into a merged count matrix ready for analysis. You can load this dataset to explore the features and workflow of ChromTag without uploading your own files.

▶ START

Data Preview

Show 10 **entries**   **Search:**

	Chromosome	Start	End	H3K27me3_rep1	H3K27me3_rep2	H3K4me3_rep1	H3K4me3_rep2
1	chr1	27805	30656	0	0	233	186
2	chr1	135727	140083	0	2	166	89
3	chr1	198335	200968	0	0	219	169
4	chr1	391014	392146	1	3	0	1
5	chr1	392617	393317	2	1	0	0
6	chr1	393808	394451	2	1	0	0
7	chr1	440860	441530	4	2	0	1
8	chr1	442100	442830	2	0	0	0
9	chr1	492154	494761	0	1	73	36
10	chr1	587572	588358	7	2	0	0

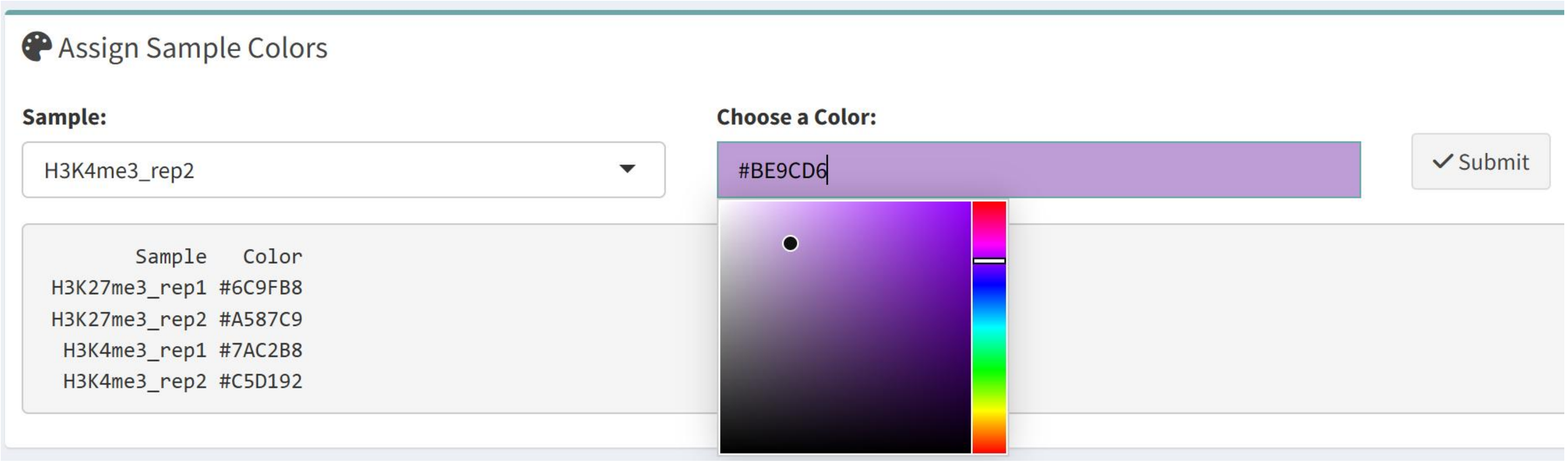
Showing 1 to 10 of 220,230 entries   Previous 1 2 3 4 5 … 22,023 Next

After choosing the data source, users can click the **"Start"** button to load the dataset and proceed to the next module.

Once the data is successfully loaded, the system will display the **peak count matrix** in a preview table for inspection.

Peaks Visualization Module

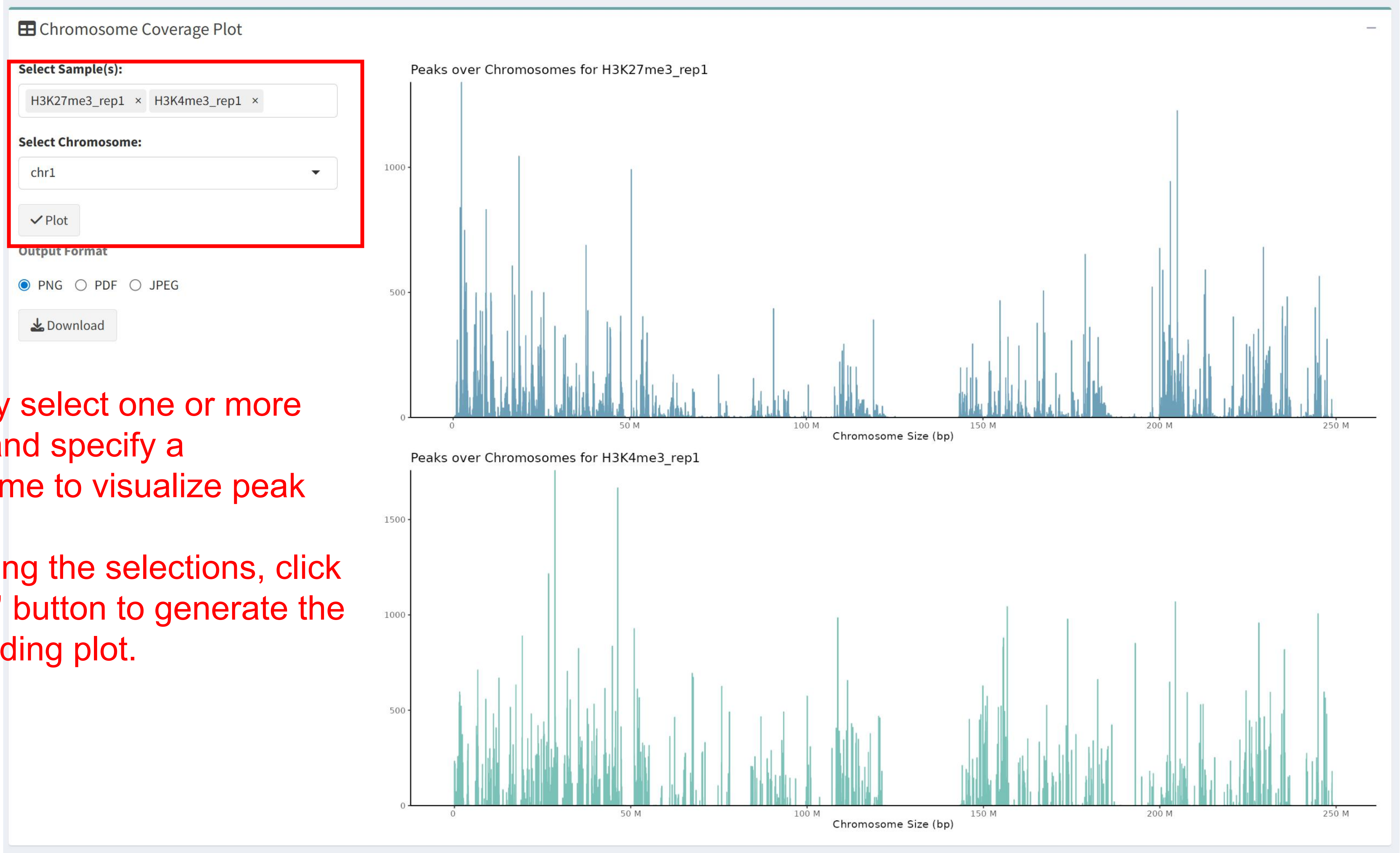
This step is optional and can be skipped.



Users can select a sample and assign a preferred color using the color picker. Once submitted, the chosen color will be applied consistently across the **Chromosome Coverage Plot** and the **TSS Profile Plot**, enhancing visual distinction among samples.

Peaks Visualization Module

This step is optional and can be skipped.

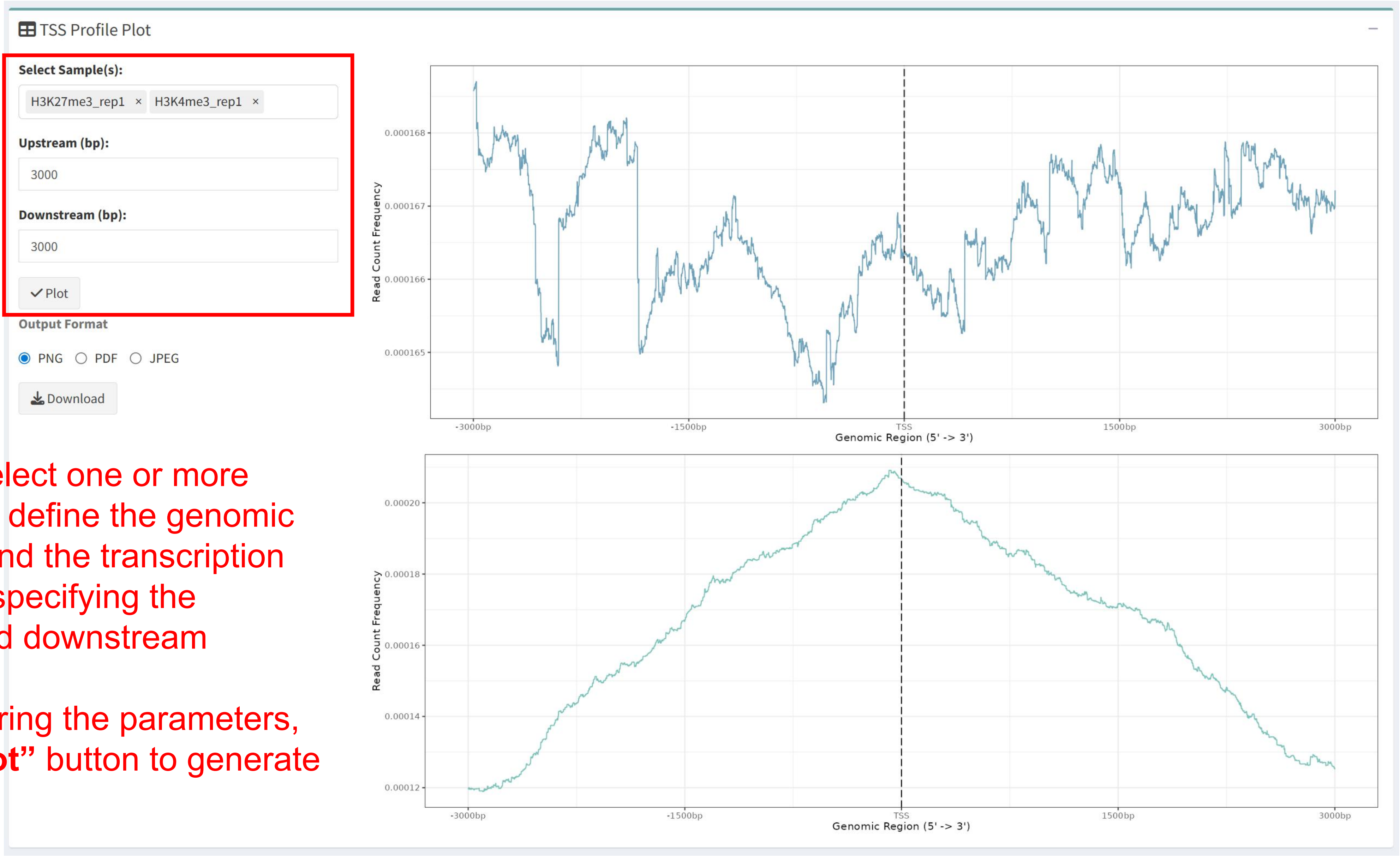


Users may select one or more samples and specify a chromosome to visualize peak coverage.

After making the selections, click the “**Plot**” button to generate the corresponding plot.

Peaks Visualization Module

This step is optional and can be skipped.



Users can select one or more samples and define the genomic window around the transcription start site by specifying the upstream and downstream distances.

After configuring the parameters, click the "**Plot**" button to generate the plot.

Data Filtering and Grouping Module

Before Filter | After Filter

Count Threshold: ?

5

✓ Submit

Users may specify a threshold value to filter the data.
During filtering, the system sums the read counts for each genomic region across samples and retains only those entries whose total count exceeds the defined threshold.

Show 5 **entries** **Search:**

hromosome	Start	End	H3K27me3_rep1	H3K27me3_rep2	H3K4me3_rep1	H3K4me3_rep2	
r1	27805	30656	0 +	0 +	233 +	186	> 5
r1	135727	140083	0	2	166	89	
r1	198335	200968	0	0	219	169	
r1	391014	392146	1	3	0	1	
r1	392617	393317	2 +	1 +	0 +	0	< 5

Showing 1 to 5 of 220,230 entries

Previous 1 2 3 4 5 … 44,046 Next

Data Filtering and Grouping Module

⇄ Sample Grouping

Select Samples:

H3K27me3_rep1 × H3K27me3_rep2 ×

Select the samples that should be grouped together.

Group Name:

H3K27me3

Enter a name for the current group.

Add Group | Clear Groups

Use the **"Add Group"** button to create the group.

Use the **"Clear Groups"** button to remove all existing groups.

Current Groups:

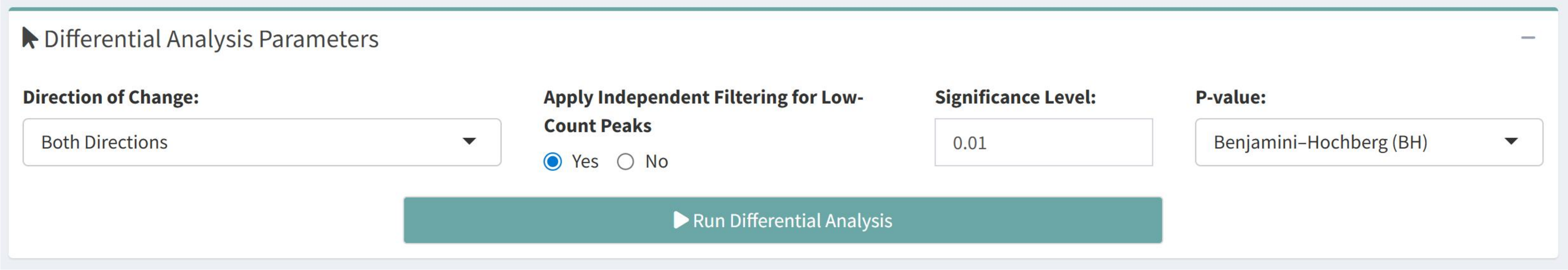
```
$H3K27me3
[1] "H3K27me3_rep1" "H3K27me3_rep2"
```

All defined groups are listed under **"Current Groups."**

✓ Submit

After all groups have been defined, click the **"Submit"** button to save the grouping information.

Differential Peak Analysis Module



Direction of Change

This setting lets users choose whether to detect peaks that are increased or decreased between groups, allowing the analysis to focus on a specific direction of change.

Independent Filtering

If you enable independent filtering, peaks with extremely low counts will be excluded. The significance level controls the cutoff applied during this process.

Significance Level

This value specifies the statistical significance threshold used when performing independent filtering.

p-value Adjustment Method

This option allows users to select a method for multiple-testing correction.

- The Benjamini–Hochberg (BH) method controls the false discovery rate (FDR) and is well suited for analyses with many tests.
- The Holm method controls the family-wise error rate (FWER) and is more conservative, reducing false positives but potentially missing true signals.

Differential Peak Analysis Module

·Comparison between two groups

Adjusted P-value Threshold:

0.05

Log2 Fold Change Threshold:

2

✓ Submit

Results can be filtered using adjusted p-values and $\log_2$ fold-change thresholds.

·Comparison between multiple groups

Differential Peak Results

1 vs 2 | 1 vs 3 | 2 vs 3

Displays multiple analysis results.

Show 10 entries

	seqnames	start	end	H3K27me3_rep1	H3K27me3_rep2
1	chr1	27805	30656	0	0
2	chr1	135727	140083	0	2

Selected Comparison:

1 vs 2

Select one comparison to use for downstream analysis.

Adjusted P-value Threshold:

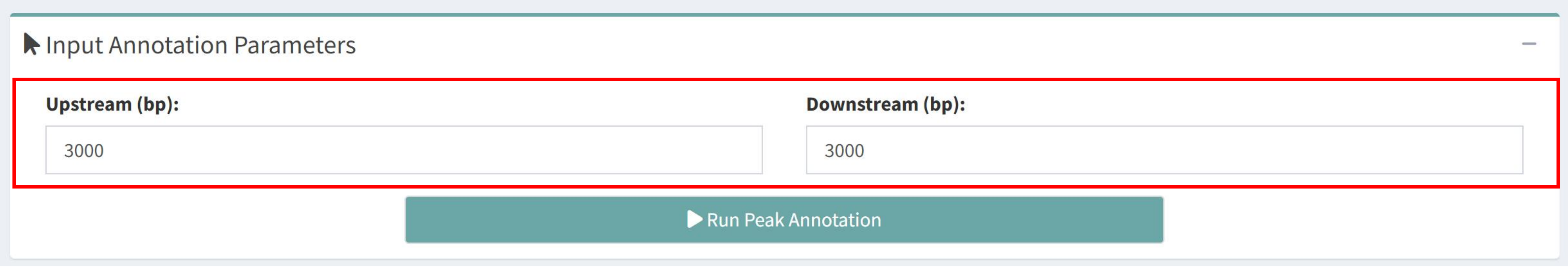
0.05

Log2 Fold Change Threshold:

2

✓ Submit

Differential Peak Annotation Module



To annotate the peaks, first specify the upstream and downstream distances to define the genomic region of interest, and then click the “Run Peak Annotation” button to execute the annotation.

Differential Peak Visualization Module

Volcano Plot | MA Plot | PCA Plot | Heatmap

–log10(p-value) Cutoff

2

Log2FC Threshold (Positive)

14

Log2FC Threshold (Negative)

5

Point Size

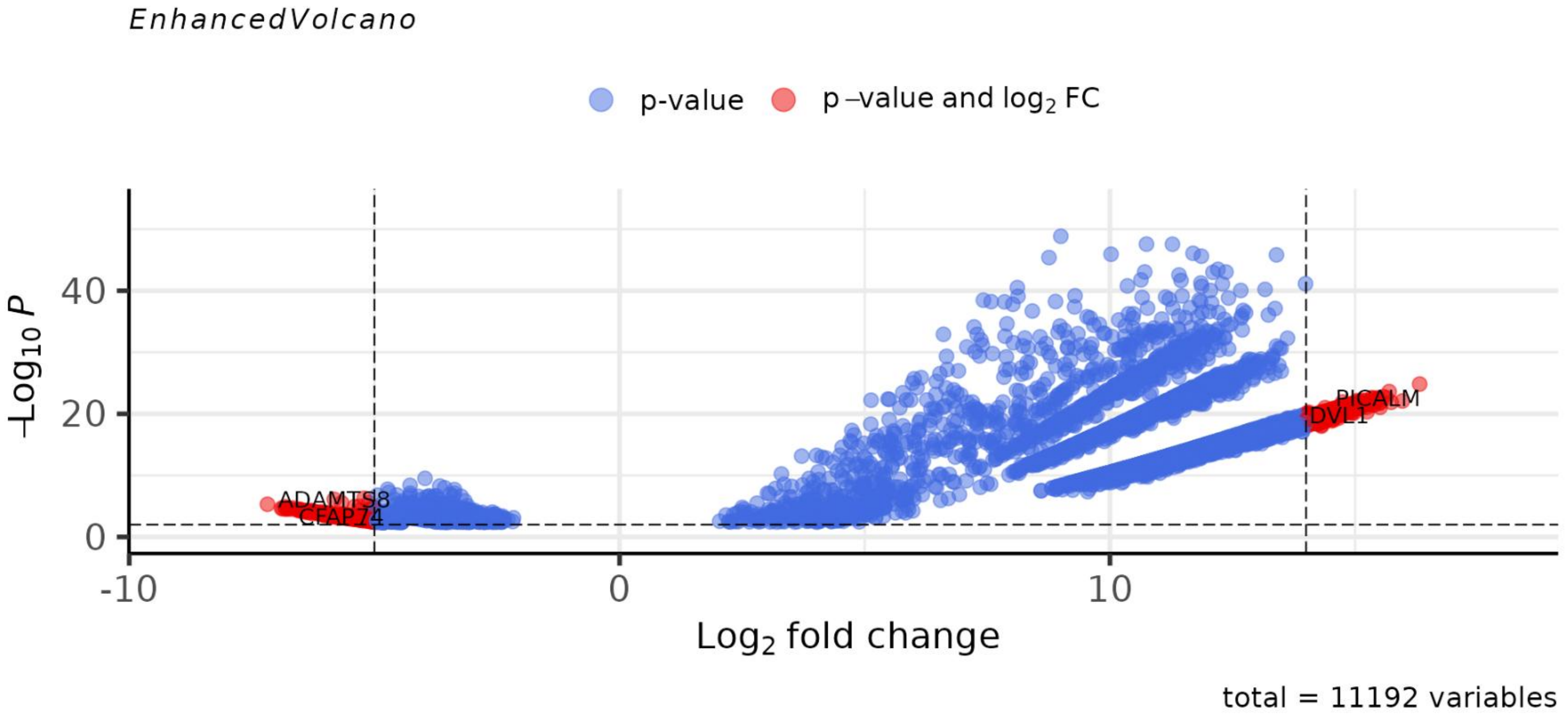
3

Label Size

4

✓ Submit

Significant peaks are identified by adjusting these three threshold settings.



Click the “Submit” button to update the plot and apply the selected significance criteria.

Differential Peak Visualization Module

Up Genes Preview

Upregulated Genes List

DVL1
MRPL20
NOL9
DNAJC11
DHRS3
RPL11
ELOA-AS1
SRRM1
RSRP1
MACO1

Show 5 entries

Search:

	seqnames	start	end	width	strand	H3K27me3_rep1	H3K27me3_rep2	H3K4me3_
17	chr1	1346468	1350600	4133	*	0	0	
20	chr1	1405327	1408434	3108	*	0	0	
77	chr1	6552669	6555289	2621	*	0	0	
82	chr1	6699924	6702352	2429	*	0	0	
142	chr1	12614231	12620067	5837	*	0	0	

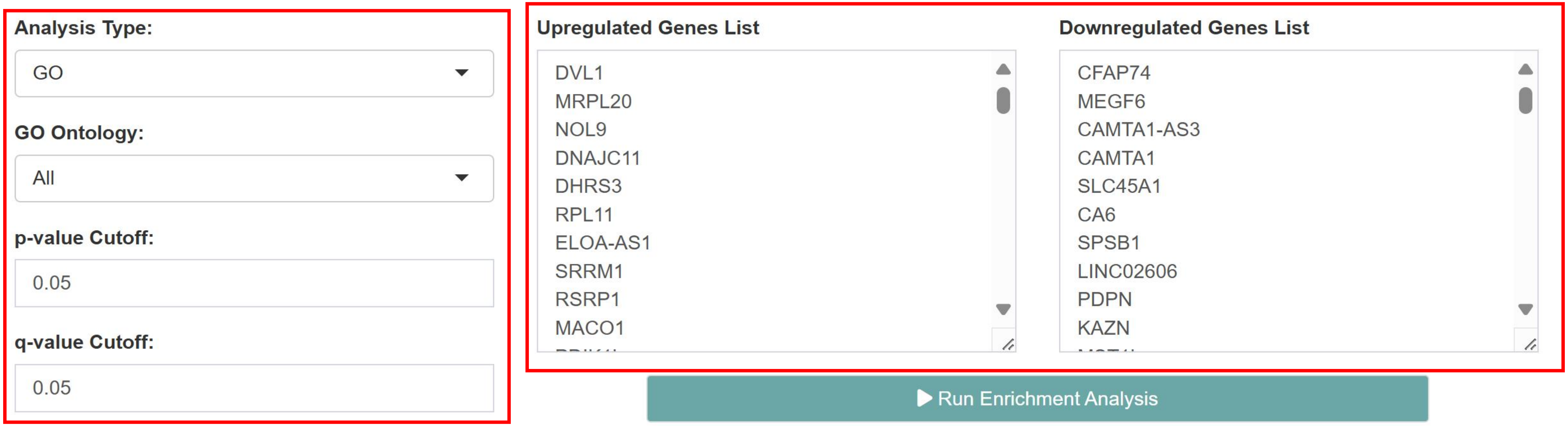
Showing 1 to 5 of 497 entries

Previous 1 2 3 4 5 … 100 Next

Download

This section displays the upregulated and downregulated gene lists, as well as the peak tables generated from the filtering step.

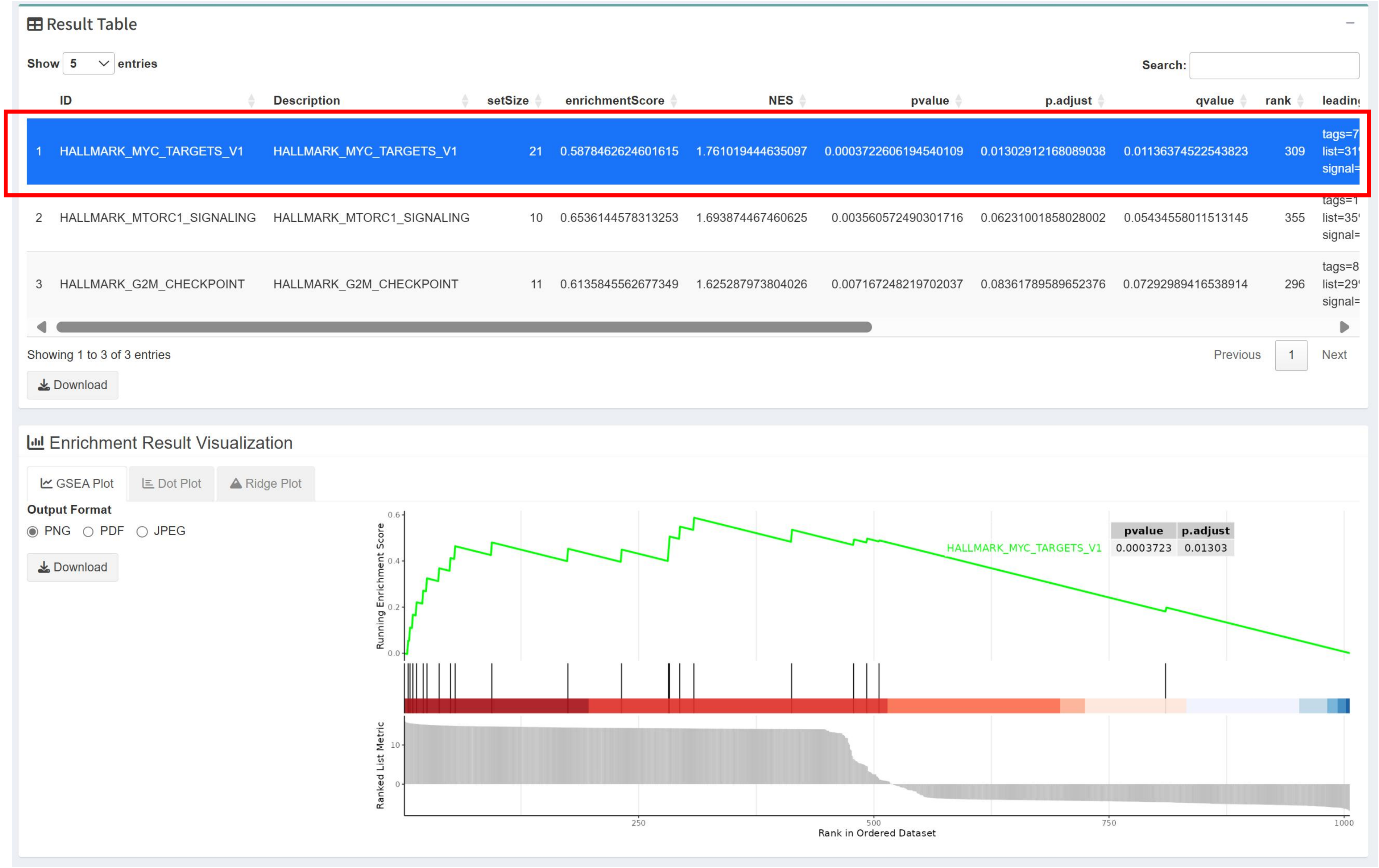
Functional Enrichment Module



Adjust the analysis type and corresponding parameters as needed.

For GO or KEGG analysis, users may enter or edit gene lists using gene symbols to define the input for enrichment testing. When GSEA is selected, the input is limited to the full set of genes annotated in the Gene Annotation module.

Functional Enrichment Module



	ID	Description	setSize	enrichmentScore	NES	pvalue	p.adjust	qvalue	rank	leadin
1	HALLMARK_MYC_TARGETS_V1	HALLMARK_MYC_TARGETS_V1	21	0.5878462624601615	1.761019444635097	0.0003722606194540109	0.01302912168089038	0.01136374522543823	309	tags=7 list=31 signal=
2	HALLMARK_MTORC1_SIGNALING	HALLMARK_MTORC1_SIGNALING	10	0.6536144578313253	1.693874467460625	0.003560572490301716	0.06231001858028002	0.05434558011513145	355	tags=1 list=35 signal=
3	HALLMARK_G2M_CHECKPOINT	HALLMARK_G2M_CHECKPOINT	11	0.6135845562677349	1.625287973804026	0.007167248219702037	0.08361789589652376	0.07292989416538914	296	tags=8 list=29 signal=

pvalue	p.adjust
0.0003723	0.01303

When GSEA is selected, users must choose a specific enrichment entry before generating the corresponding GSEA plot.

Motif Enrichment Module

Upregulated Peaks List

chr3-150407290-150415877
chr16-2150741-2157314
chr16-88972494-88978084
chr6-119343897-119351533
chr12-56682168-56690455
chr8-38173859-38178882
chr9-128688198-128692391
chr11-65420641-65428391
chr10-92688412-92694886
chr6-27889944-27897392

Downregulated Peaks List

chr7-154746354-154754198
chr11-20155235-20165969
chr7-27100472-27112603
chr1-208072367-208087339
chr1-8323134-8331215
chr11-118150179-118161808
chr20-47665285-47675999
chr14-64734713-64750256
chr1-28810214-28820062
chr4-182814697-182830407

This section displays the top 200 upregulated and 200 downregulated peaks, which are selected from the differential peaks identified through the volcano plot filtering step. You can also input and modify your own lists of upregulated and downregulated peaks for further analysis.

Number of Top Transcription Factors to Plot

10

Display Motif GC Content

- (•) Yes
- ( ) No

Enable Clustering

- (•) Yes
- ( ) No

·**Number of Top Transcription Factors to Plot**: Set how many top transcription factors you want to visualize (default is 10).
·**Display Motif GC Content**: Choose whether to show the GC content of motifs.
·**Enable Clustering**: Decide whether to enable clustering for grouping similar motifs in the plot.